Form D

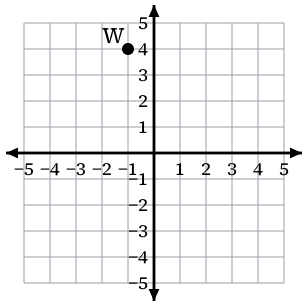
Core Focus on Proportions & Probability

Block 5 Review ~ Direct Variation

Name _____ Period _____ Date _____

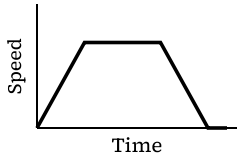
Part I – Selected Response

1. Which ordered pair represents the point graphed?



A. (4, −1)
B. (1, −4)
C. (−4, 1)
D. (−1, 4)

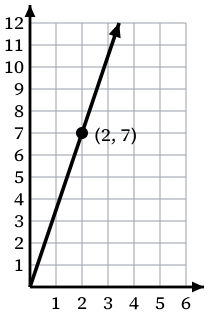
2. Which story best matches the speed graph below?



A. Ella walked to the park, sat on a bench, and then walked home.
B. A runner ran at the same speed the whole time.
C. A skateboarder slowed down, stopped, and then sped up again.
D. A bus sped up, drove at a steady speed, then slowed down and stopped at a bus stop.

3. If the graph below models direct variation, what is its rate?

A. $\frac{2}{7}$
B. 3.5
C. 7
D. The graph does not model direct variation.

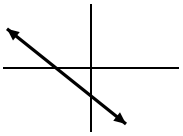
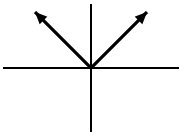
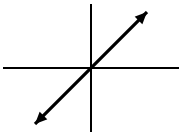
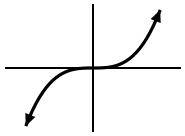
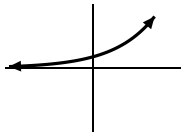
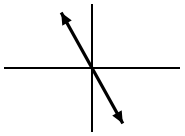


4. Which equation is the direct variation equation for the values in the table below?

x	4	8	12	16
y	3	6	9	12

A. y = 3x
B. $y = \frac{4}{3}x$
C. $y = \frac{3}{4}x$
D. y = x − 1

5. Which graphs show direct variation? Circle all that apply.

A. 
B. 
C. 
D. 
E. 
F. 

6. Which equations are direct variation equations? Circle all that apply.

A. y = 8 + x
B. y = x
C. $y = \frac{12}{x}$
D. y = −2.5x
E. y = 4 − 3x
F. $y = -\frac{x}{2}$

7. The table below models direct variation. What is the missing value?

x	0	2	4	...	9
y	0	5	10	...	?

A. 18

B. 19

C. 22.5

D. 45

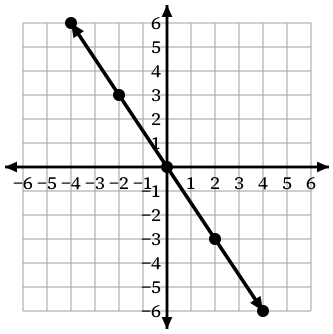
8. What is the direct variation equation for the line graphed below?

A. $y = -\frac{3}{2}x$

B. $y = -\frac{2}{3}x$

C. $y = \frac{3}{2}x$

D. $y = -3x$



Form D

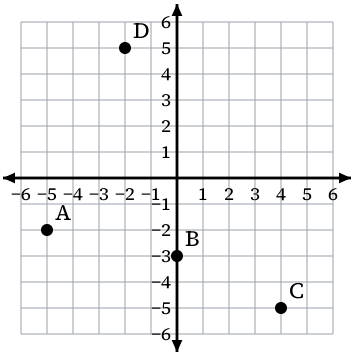
Core Focus on Proportions & Probability

Block 5 Review ~ Direct Variation

Name _____ Period _____ Date _____

Part II – Constructed Response

Write the ordered pair for each point on the coordinate plane to the right. Then name the quadrant the point is in, or the axis it is on.



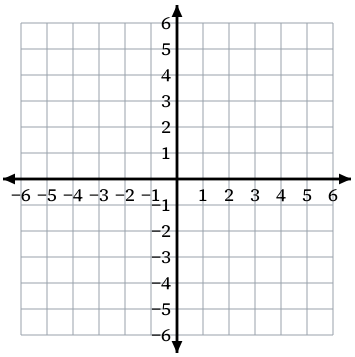
1. A

2. B

3. C

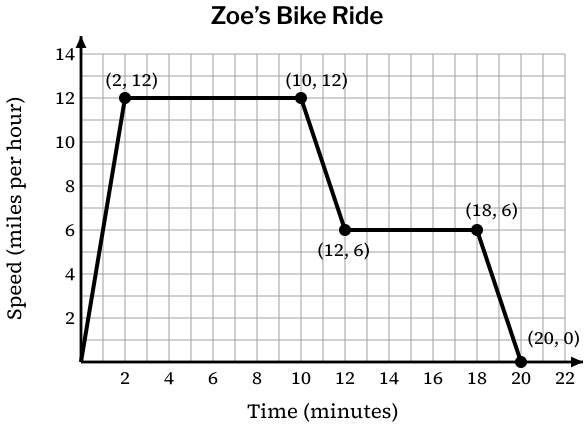
4. D

5. The points A(−5, 4), B(1, 4) and C(1, −2) are three corners of square ABCD. Plot A, B and C. Then plot the fourth corner, D, and connect the points in order to make the square. What are the coordinates of D?



D = _____

6. Zoe rode her bike to a friend's house. The graph below shows her **speed** (miles per hour) over time (minutes). For each part, explain how you know your answer is correct.



a. What was Zoe's fastest speed?

b. For how many minutes did Zoe ride at her fastest speed?

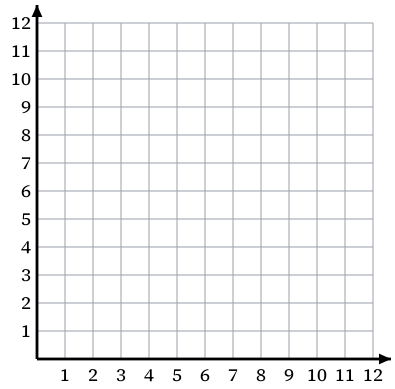
c. What was happening to Zoe's speed from minute 10 to minute 12?

d. What was Zoe doing from minute 12 to minute 18? Was she stopped?

Complete each input-output table and graph each function in Quadrant I. What is the slope of each direct variation equation?

7.

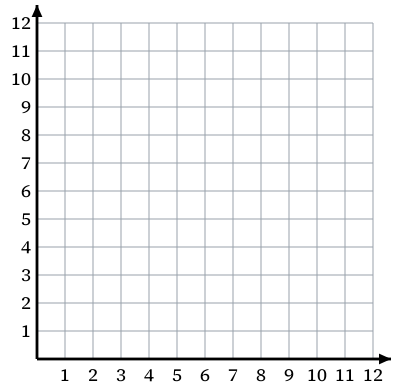
Input x	Function Rule y = 2.5x	Output y
0		
1		
2		
3		
4		



slope: _____

8.

Input x	Function Rule $y = \frac{3}{4}x$	Output y
0		
4		
8		
12		



slope: _____

Determine whether each table models direct variation. Circle YES or NO. If it does, give the rate. If not, explain why not.

9.

x	y
−4	−10
−2	−5
2	5
4	10
6	15

direct variation? **YES** **NO**

rate/explanation:

10.

x	y
−3	9
−1	1
0	0
2	4
4	16

direct variation? **YES** **NO**

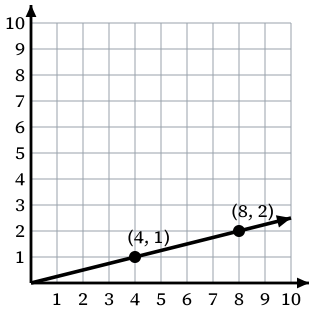
rate/explanation:

Determine whether each equation models direct variation. Circle YES or NO. If it does, give the rate. If not, explain why not.

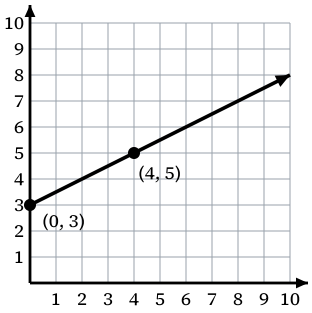
11. $y = -x$ direct variation? **YES** **NO** rate/explanation:

12. $y = \frac{x}{2} + 1$ direct variation? **YES** **NO** rate/explanation:

13. Omar says Graph A models direct variation and has a rate of 4, because $8 \div 2 = 4$. Priya says Graph B models direct variation because it is a straight line and has a rate of $\frac{1}{2}$. Is either student correct? Explain each student's mistake.



Graph A

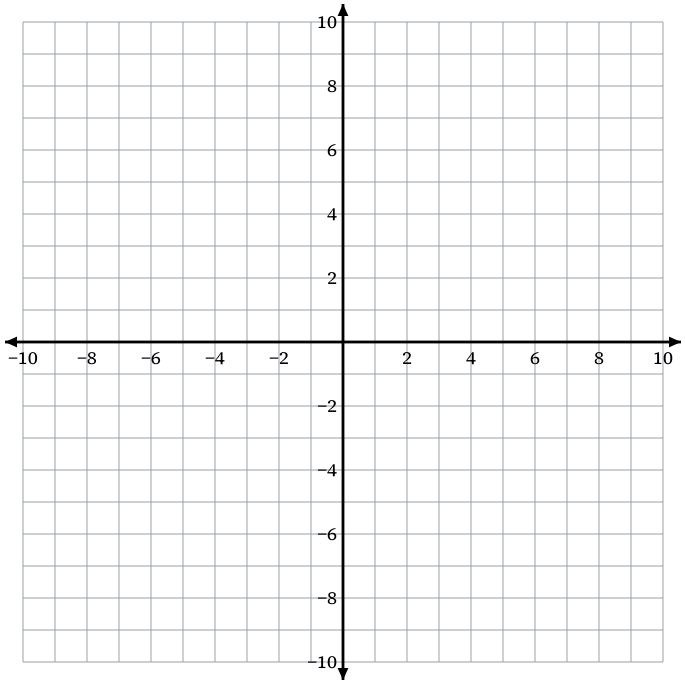


Graph B

14. A machine fills juice bottles at a steady rate. The table below shows the number of bottles filled, y, after x minutes. Write a direct variation equation for the table. Then use it to find how many bottles the machine fills in 25 minutes.

x	y
2	30
5	75
8	120
10	150

15. Graph the direct variation equation $y = -\frac{2}{3}x$.



Core Focus on Proportions & Probability

Block 5 Review ~ Direct Variation

Name _____ Period _____ Date _____

Part III – Problem Solving

The total cost of bags of mulch varies directly with the number of bags. The table below shows what Theo paid for mulch for his garden.

Bags of Mulch	Total Cost ($)
0	0.00
4	27.00
8	54.00
12	81.00
16	108.00

How many bags of mulch can Theo buy for $148.50?